Gap Analysis of AI Agent Protocols

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Motivations

- Cross-domain AI Agent interaction use cases are emerging, and it's IETF's role to take care.
- It's important to define AI Agent protocols that are **interoperable**, **vendor-neutral**, and **scalable**.
- Use cases taxonomy, terminologies consensus, and **Gap Analysis** are essential before defining solutions.

Problem Space

- AI Agent Identity, Authentication, and Authorization

  (will not be covered in this side meeting)

- AI Agent Discovery

  (part 2 discussion)

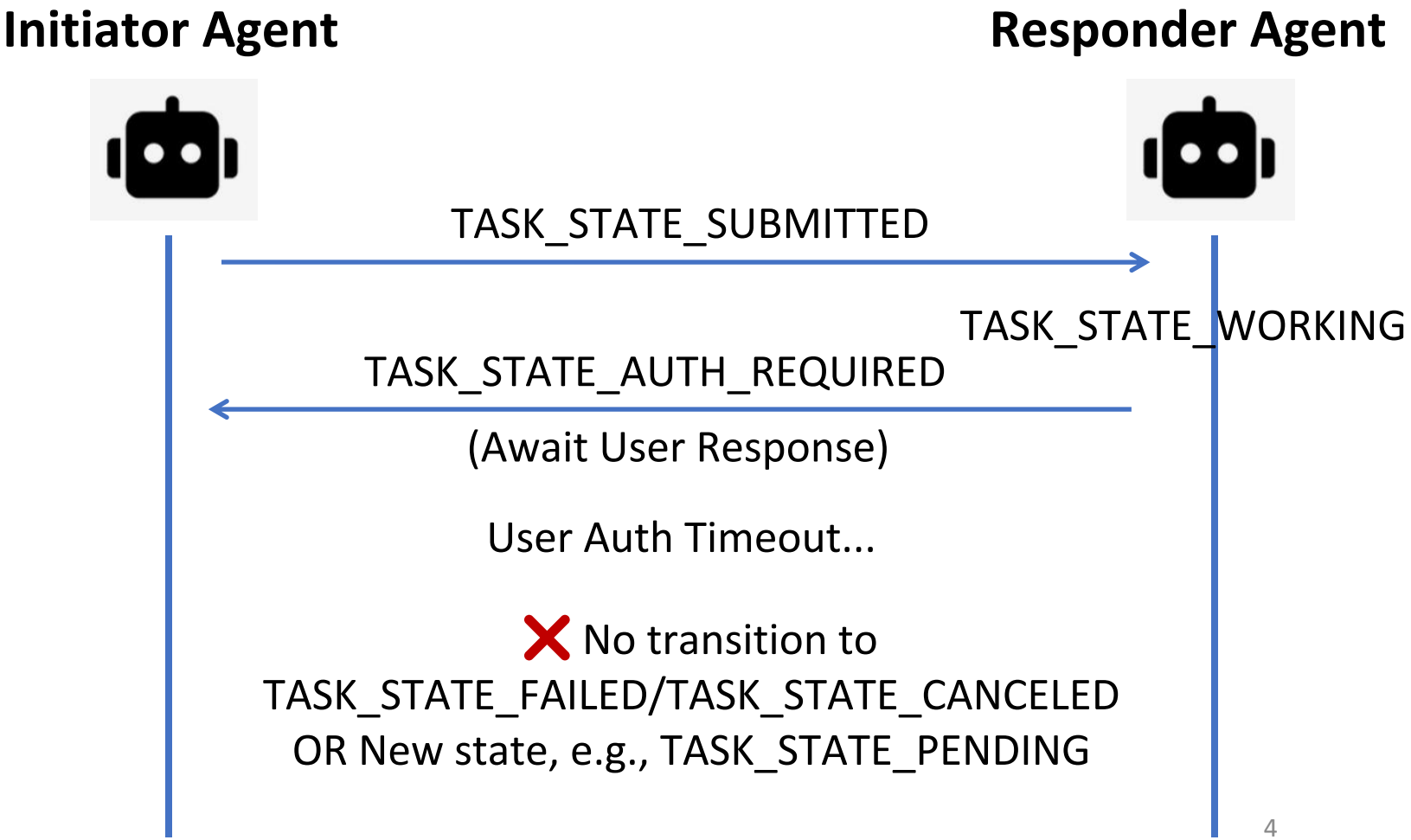
- **Session Management**
- **Multi-modal Transport**

(This Slides)

Session Management

- **Issue-1:** A2A defines 8 interactive states. If the transition rules between these states are not carefully specified, abnormal state flows are highly likely to occur.

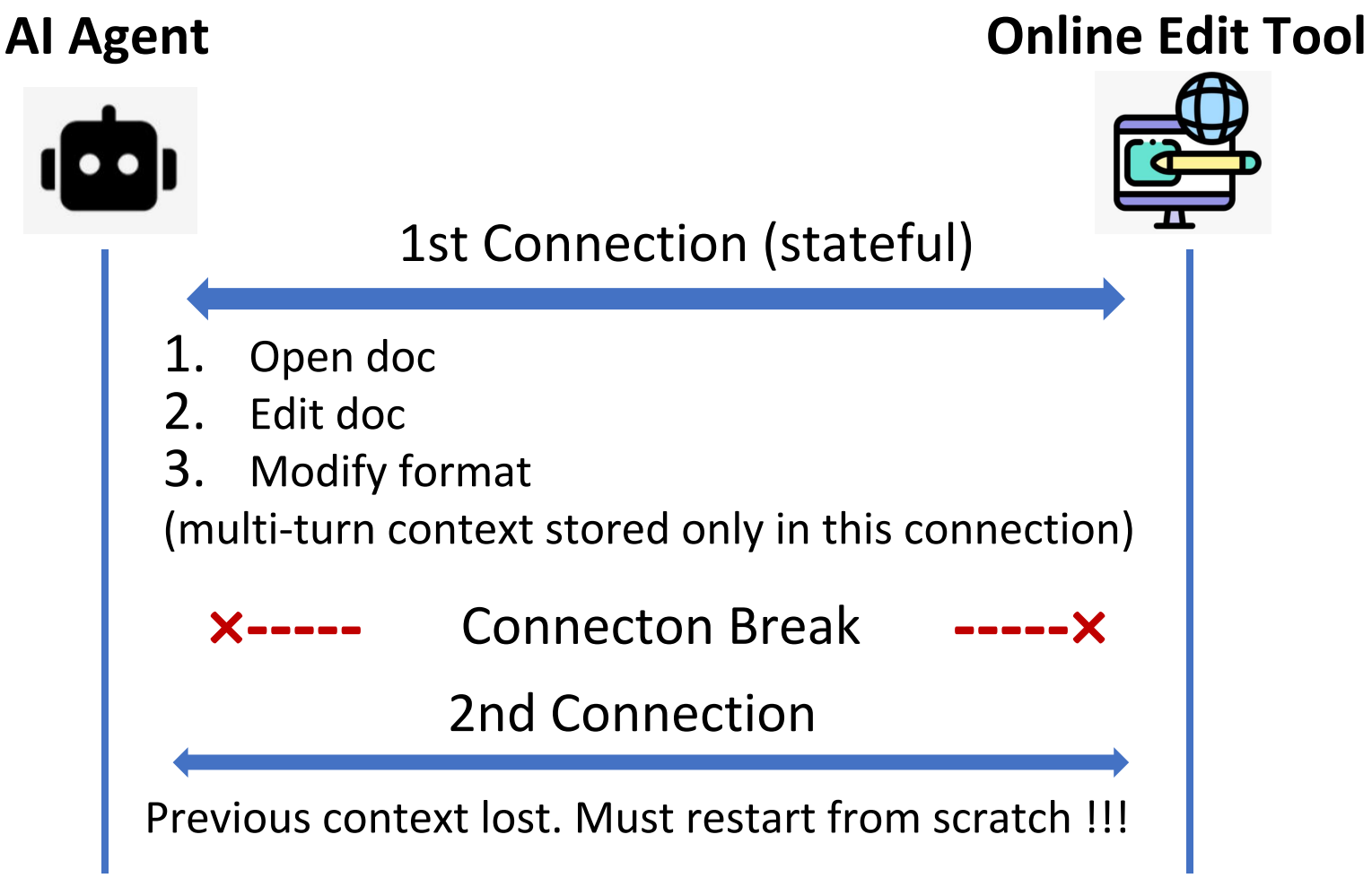
- v1.0: `"TASK_STATE_SUBMITTED"`, `"TASK_STATE_WORKING"`, `"TASK_STATE_COMPLETED"`, `"TASK_STATE_FAILED"`, `"TASK_STATE_CANCELED"`, `"TASK_STATE_REJECTED"`, `"TASK_STATE_INPUT_REQUIRED"`, `"TASK_STATE_AUTH_REQUIRED"`



Session Management

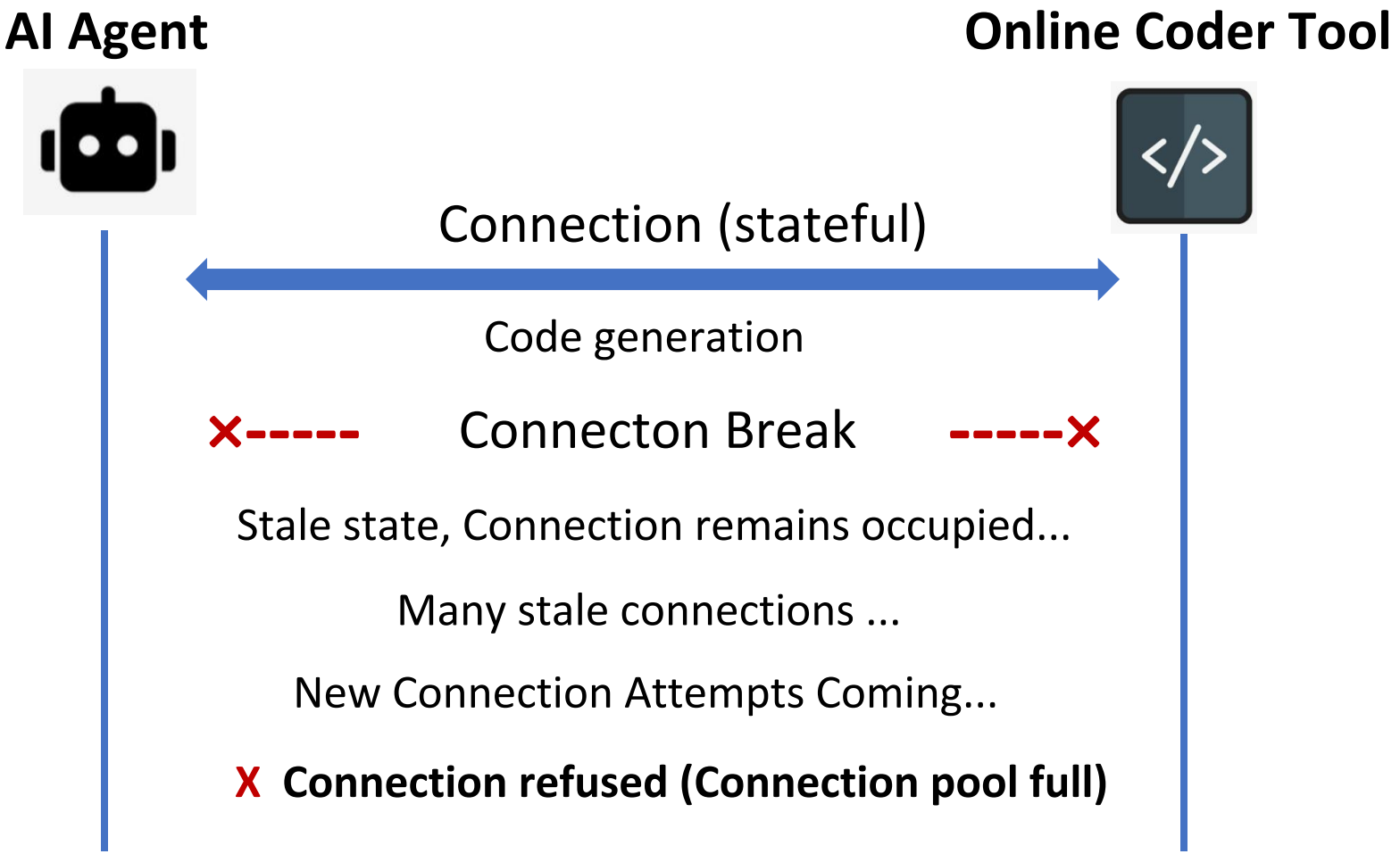
- **Issue-2:** MCP is stateful per connection and stateless across connections. It does not support session resumption.

- **Example Case:**



Session Management

- **Issue-3:** MCP does not support native session timeouts. Applications can retry but this is implementation-specific.
- **Example Case:**

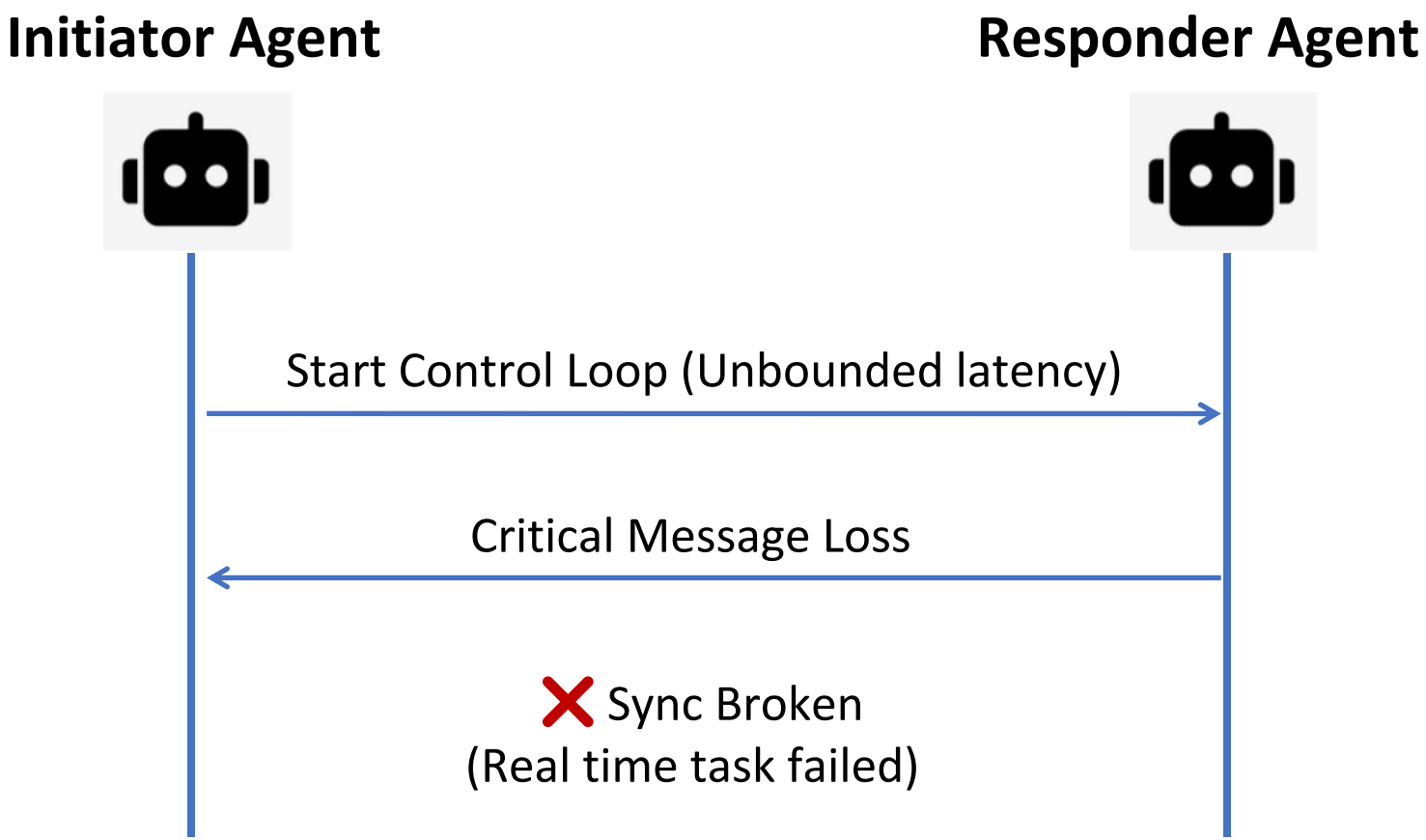


Session Management

- A good AI Agent session protocol Should consider:
  - Session Timeout and Expiration
  - Session Recovery and Reconnection
  - Session State Semantics
  - Session level constrains

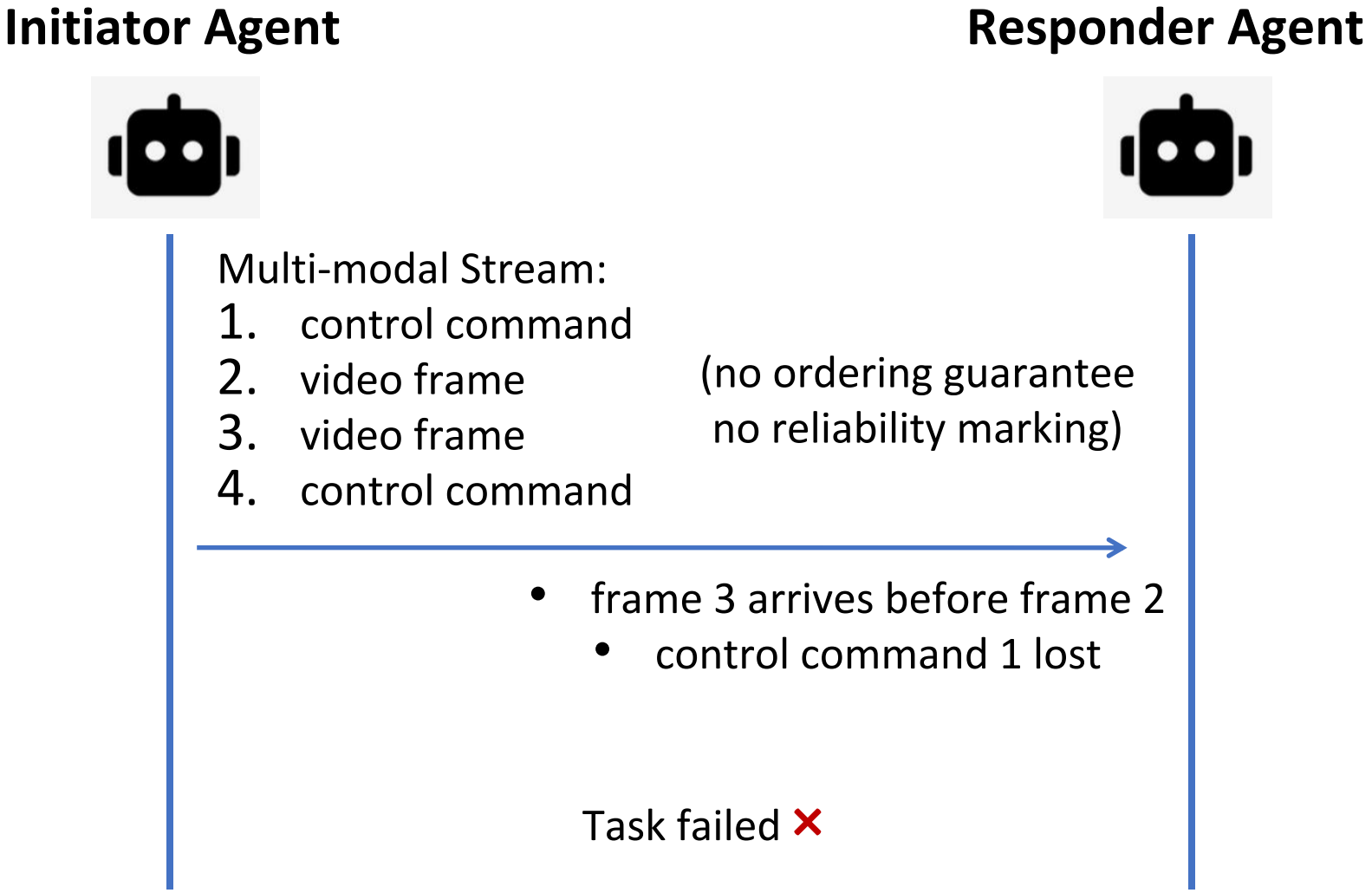
Multi-modal Transport

- **Issue-1:** Lack mechanism for **predictable**/controlled ordering and **loss handling** of task for critical agent message. As a result, timing-sensitive agent behaviors cannot rely on predictable inter-agent message timing.



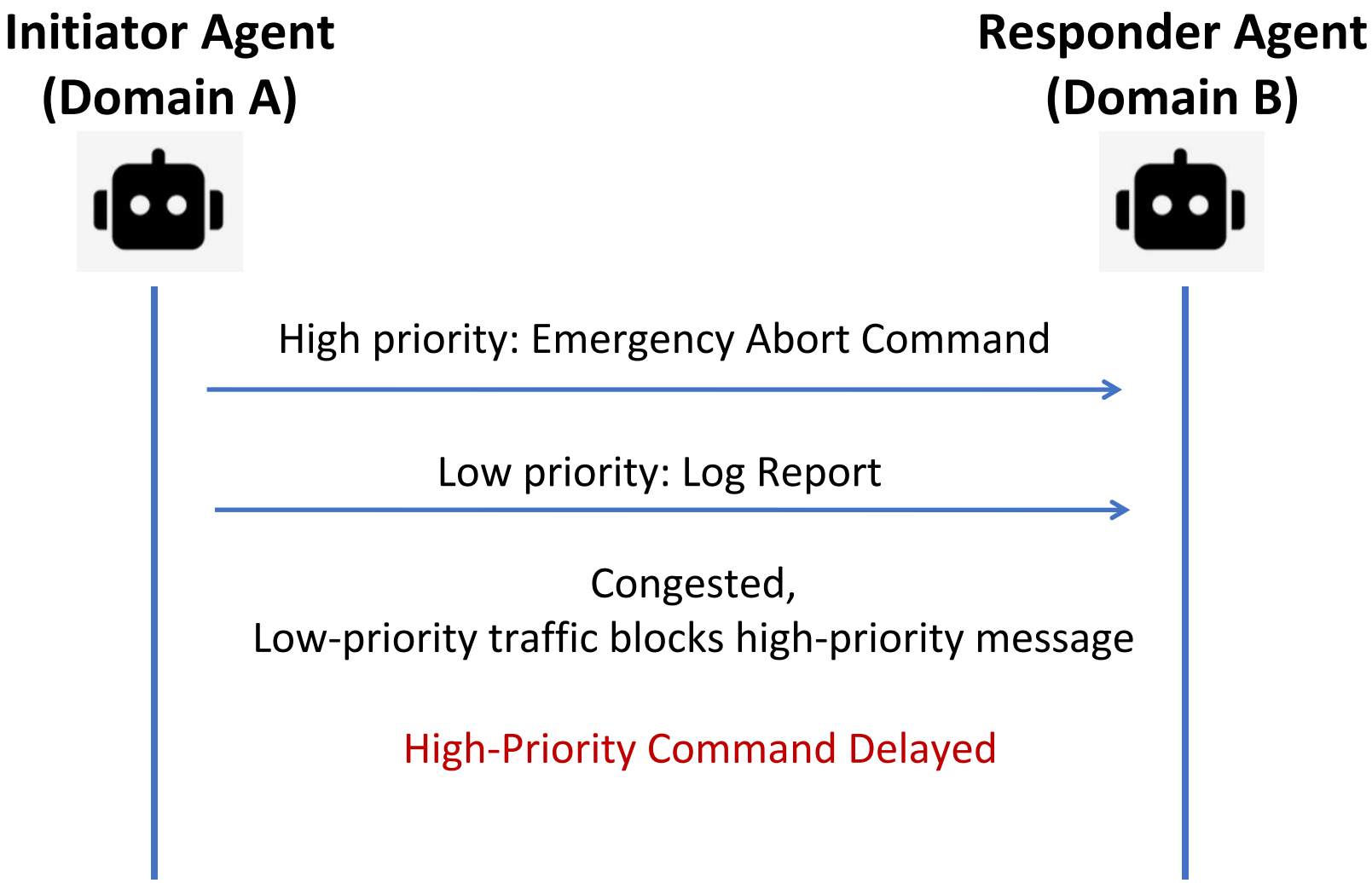
Multi-modal Transport

- **Issue-2:** MCP and A2A both lack mechanisms for conveying **message ordering** requirements across multi-modal data transmission. Similarly, no mechanisms exists to distinguish between messages that e.g. must be reliably delivered, those that may be dropped or superseded.



Multi-modal Transport

- **Issue-3:** both MCP and A2A leave **prioritization and scheduling** of message and task completion to the application, **not in the protocol**, preventing cross-domain agents from guaranteeing priority processing for high-priority messages.



Multi-modal Transport

- Mapping agent interactions onto multi-modal transport protocols Should consider:
  - Latency bounded transmission
  - Message Ordering
  - Loss handling
  - Priority scheduling